

MEDINI CRM — EXTERNAL SYSTEM STUDY & GAP ANALYSIS

The Dr Partner
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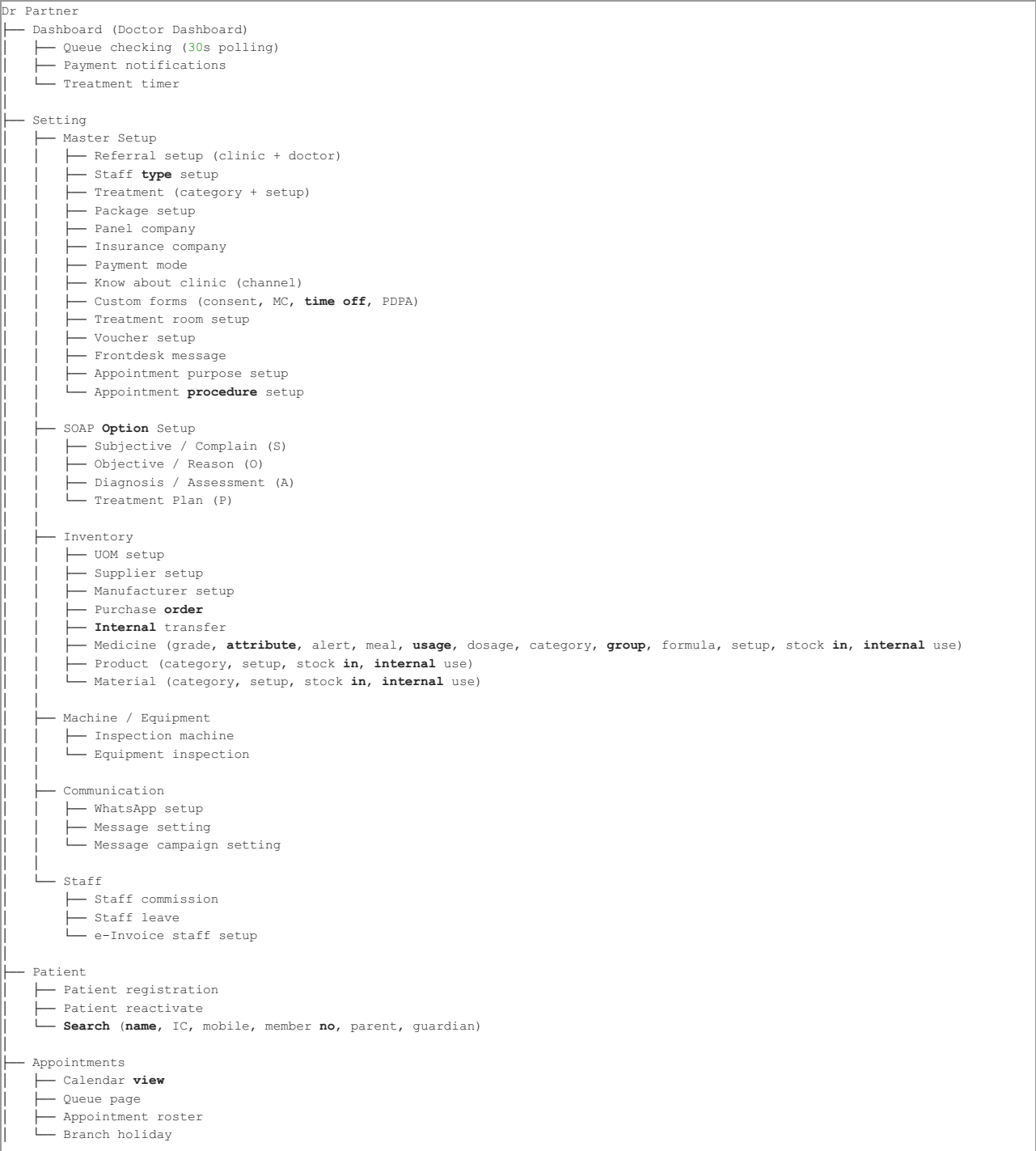
1. Executive Summary

The Dr Partner is a comprehensive clinic management system used by Medini Dental Group. It covers patient management, appointments, treatment, payments, insurance, inventory, reporting, and branch operations. This study identifies what Medini CRM currently has, what is missing, and what should be prioritized before production backend development.

Key Finding: The Dr Partner is a mature, feature-rich system with deep operational workflows. Medini CRM has a superior dashboard/intelligence layer but lacks the depth in patient financial management, treatment catalog, insurance/panel workflows, and clinical documentation.

2. System Overview

2.1 Module Map (Actual Structure Observed)



Point of Sale (POS)	POS	POS void
Payment	Payment collection	Refund payment
	Void payment	
Insurance / Panel	Panel invoice	Panel payment
	Panel void invoice	Insurance invoice
	Insurance payment	Insurance void invoice
Outstanding / Credit	Outstanding invoice	Unpaid invoice
	Branch credit setup	
Reports	Daily patient visit	Monthly patient visit
	Monthly visit summary	Yearly patient visit
	Daily patient feedback	Monthly patient feedback
	New patient report	Tourism report
	Daily treatment	Monthly treatment
	Yearly treatment	Patient last treatment
	Daily medicine	Monthly medicine
	Daily product	Monthly product
	Daily material	Monthly material
	Deposit report	Incident report
	Death report	DOSH inspection
Branch Management	Branch edit	Branch logo
	Branch staff setup	Branch tax edit
	Branch credit setup	

3. Patient Management

3.1 Patient Fields Observed

Field	Purpose	Required	Medini Equivalent	Recommendation
Name	Patient identity	Yes	☒ Exists	—
IC No	Identity verification	Yes	☒ Exists (Phase 6.3)	—
Mobile No	Contact	Yes	☒ Exists	—
Member No	Membership tracking	Optional	☒ Missing	P2 — Add member/loyalty ID
Parent Name	Family linkage	Optional	☒ Exists (Phase 6.1)	—
Parent Phone	Family contact	Optional	☒ Exists (Phase 6.3)	—
Guardian Name	Legal guardian	Optional	☒ Exists (Phase 6.3)	—
Guardian Phone	Guardian contact	Optional	☒ Exists (Phase 6.3)	—
DOB	Age calculation	Yes	☒ Exists	—
Gender	Demographics	Yes	☒ Exists	—
Address	Location	Optional	☒ Missing	P1 — Add address field
Email	Digital contact	Optional	☒ Missing	P2 — Add email field
Occupation	Demographics	Optional	☒ Missing	P3 — Add occupation
Race/Ethnicity	Demographics	Optional	☒ Missing	P3 — Add if needed for MOH reporting
Allergies	Clinical safety	Optional	☒ Missing	P1 — Critical for clinical safety
Medical History	Clinical context	Optional	☒ Missing	P1 — Important for treatment planning
Dental History	Clinical context	Optional	☒ Missing	P1 — Important for continuity
Notes	General notes	Optional	☒ Exists	—

3.2 Critical Missing Patient Fields

P0 — Critical:

- **Allergies** — Clinical safety requirement. Must be visible during treatment.
- **Medical History** — Contraindications for dental procedures.
- **Address** — Required for billing, insurance, and recall letters.

P1 — Important:

- **Email** — Digital communication channel.
- **Occupation** — Demographic analysis.

P2 — Useful:

- **Member No** — Loyalty program integration.
- **Emergency Contact** — Separate from guardian for adults.

4. Treatment / Service Catalog

4.1 Treatment Categories Observed

Category	Examples
Consultation	Initial consult, follow-up consult
Preventive	Scaling, polishing, fluoride, sealant
Restorative	Filling (composite, amalgam, GIC), inlay, onlay
Endodontic	Root canal (anterior, premolar, molar), pulpotomy, pulpectomy
Prosthodontic	Crown (PFM, zirconia, E-max), bridge, denture (partial, full), implant crown
Orthodontic	Braces (metal, ceramic, self-ligating), aligner, retainer
Oral Surgery	Extraction (simple, surgical), wisdom tooth, biopsy
Periodontic	Deep scaling, root planing, gum surgery
Cosmetic	Whitening (in-office, take-home), veneer, bonding
Diagnostic	X-ray (IOPA, OPG, CBCT), study model, photo
Emergency	Pain relief, abscess drainage, temporary filling

4.2 Complete Treatment Catalog (Built-in + Custom)

Built-in treatments (system default): 48

No	Treatment	Notes
1	ALVEOLOPLASTY	Surgical ridge preparation
2	AMALGAM FILLING	Metal filling
3	BLANC ONE WHITENING	Whitening brand
4	CBCT	3D imaging
5	COMPOSITE FILLING	Tooth-colored filling
6	CONSULTATION	Doctor consult
7	DENTURE	General denture
8	DRY SOCKET MX	Dry socket management
9	EXTRACTION	Tooth removal
10	F/F DENTURE ACRYLIC	Full denture acrylic
11	F/F DENTURE FLEXIBLE	Full denture flexible
12	FILLING	General filling
13	FISSURE SEALANT	Preventive sealant
14	FLUORIDE APPLICATION	Fluoride treatment
15	FULL METAL CROWN	Metal crown
16	GIC FILLING	Glass ionomer filling
17	IMPLANT CROWN ISSUED	Implant crown delivery
18	IMPLANT IMPRESSION	Implant impression
19	INVISALIGN	Clear aligner
20	INVISALIGN MODERATE	Moderate aligner case
21	INVISALIGN REVIEW	Aligner review
22	LASER EMUNDO	Laser treatment
23	LASER GINGIVECTOMY	Laser gum surgery
24	MFT + EXPANDER	Myofunctional therapy
25	MFT + TRAINER	Trainer appliance
26	MFT REVIEW	MFT review
27	MILK TOOTH EXTRACTION	Pediatric extraction
28	OPG	Panoramic X-ray
29	ORTHO DEBOND	Braces removal
30	ORTHO REVIEW	Braces review
31	PA RADIOGRAPH	Periapical X-ray
32	PARTIAL DENTURE ACRYLIC	Partial denture acrylic
33	PARTIAL DENTURE THERMOSENSE	Flexible partial denture
34	PFM BRIDGE	Porcelain-fused-metal bridge
35	PFM CROWN	Porcelain-fused-metal crown
36	PULPOTOMY	Pulp treatment
37	RETAINER	Orthodontic retainer
38	ROOT CANAL - ANTERIOR	Front tooth RCT
39	ROOT CANAL - MOLAR	Back tooth RCT
40	ROOT CANAL - PREMOLAR	Premolar RCT
41	SCALING AND POLISHING	Cleaning
42	STO	(abbreviation — likely study model)

43 STRAUMANN IMPLANT	Implant brand
44 WHITENING	General whitening
45 WISDOM TOOTH EXTRACTION	Wisdom tooth removal
46 WISDOM TOOTH SURGERY	Surgical wisdom tooth
47 ZIRCONIA BRIDGE	Zirconia bridge
48 ZIRCONIA CROWN	Zirconia crown

Branch-custom treatments (Medini Dental Group): 21

No	Treatment
1	APPLIANCE
2	BRACES / ORTHO
3	BRIDGE
4	CBCT X-RAY
5	CLEAR ALIGNER
6	CONSULTATION
7	CROWN
8	DENTURE
9	FLOURIDE
10	GINGIVECTOMY
11	IMPLANT
12	LA CEPH XRAY
13	LOCAL ANESTHESIA
14	MINOR ORAL SURGERY
15	OPG XRAY
16	PA XRAY
17	ROOT CANAL TREATMENT / ENDO
18	SCALING & POLISHING
19	STO
20	VENEER
21	WHITENING

Note: Branch prices are set to 0.00 in the demo view — actual pricing is configured per branch in production use.

4.3 Treatment Structure Observed

Attribute	Observed	Medini Status
Treatment name	☒ Yes	☒ Exists
Treatment code	☒ Yes (item code)	☒ Missing
Category	☒ Yes	☒ Missing
Price	☒ Yes (branch-specific)	☒ Missing
Price 2	☒ Yes (alternate pricing)	☒ Missing
Panel price	☒ Yes (insurance/panel rate)	☒ Missing
Cost	☒ Yes (clinic cost)	☒ Missing
Tax % (SST)	☒ Yes	☒ Missing
Dental chart symbol	☒ Yes	☒ Missing
X-ray indicator	☒ Yes	☒ Missing
e-Invoice classification	☒ Yes	☒ Missing
Keyboard shortcut	☒ Yes	☒ Missing
Package eligible	☒ Yes	☒ Missing
Insurance eligible	☒ Yes	☒ Missing
Specialist required	☒ Yes	☒ Missing

Gap: Medini CRM needs a complete treatment catalog with codes, categories, pricing, and insurance eligibility flags.

5. Insurance / Panel

5.1 Panel/Insurance Providers Observed

The system supports panel companies and insurance companies as separate entities:

Type	Workflow
Panel Company	Direct billing to employer/corporate panel
Insurance Company	Claim submission to insurance provider

5.2 Insurance/Panel Workflow



5.3 Key Insurance Fields

Field	Purpose
Panel/Insurance name	Provider identity
Policy/member no	Patient coverage ID
Coverage limit	Maximum claimable
Patient portion	Co-payment amount
Panel portion	Billable to panel
Claim reference	Submission tracking
Claim status	Pending/Approved/Rejected
Approved amount	Final approved value
Rejected amount	Disputed value

Gap: Medini CRM has no insurance/panel module. This is P0 for real clinic operations.

6. Payment System

6.1 Payment Methods Observed

Method	Observed
Cash	☒ Yes
Card (Visa/Master)	☒ Yes
Online Transfer	☒ Yes
E-Wallet	☒ Yes (implied)
Insurance	☒ Yes
Panel	☒ Yes
Deposit	☒ Yes
Credit	☒ Yes
Cheque	Not observed

6.2 Payment Features

Feature	Observed	Medini Status
Full payment	☒ Yes	☒ Exists
Partial payment	☒ Yes	☒ Missing
Deposit	☒ Yes	☒ Missing
Refund	☒ Yes	☒ Missing
Void payment	☒ Yes	☒ Missing
Outstanding tracking	☒ Yes	☒ Missing
Receipt generation	☒ Yes	☒ Missing
Invoice generation	☒ Yes	☒ Missing

Gap: Medini CRM has no payment/invoice/outstanding system. This is P0.

7. Partial Payment / Outstanding Model

7.1 Observed Model

```
Treatment Value: RM1,000
↓
Patient Pays: RM500 (Cash) + RM300 (Panel)
↓
Outstanding: RM200
↓
Status: Partially Paid
```

7.2 Financial States Observed

State	Description
Unpaid	No payment received
Partially Paid	Some payment, balance remaining
Paid	Full payment received
Refunded	Payment returned
Void	Transaction cancelled
Outstanding	Balance due
Deposit	Advance payment held

7.3 Outstanding Invoice Fields

Field	Purpose
Invoice date	Transaction date
Patient name	Debtor
Treatment	Service rendered
Total amount	Full charge
Paid amount	Amount received
Outstanding balance	Amount due
Status	Payment state

Gap: Medini CRM has no partial payment or outstanding tracking. This is P0.

8. Appointment Workflow

8.1 Statuses Observed

Status	Description
Booked	Appointment scheduled
Confirmed	Patient confirmed
Checked-in	Patient arrived
Waiting	In queue
In Progress	Being treated
Completed	Treatment done
Cancelled	Appointment cancelled
No-show	Patient did not arrive

8.2 Queue Management

- Real-time queue checking (30-second polling)
- Treatment timer
- Payment notification
- Queue page display

Medini Status: Basic appointment exists (Phase 6.2). Queue management missing.

9. Clinical Workflow (SOAP)

9.1 SOAP Structure Observed

Component	Description
S — Subjective	Patient complaint
O — Objective	Clinical findings
A — Assessment	Diagnosis
P — Plan	Treatment plan

9.2 Clinical Documentation

Feature	Observed
Chief complaint	☒ Yes
Diagnosis	☒ Yes
Treatment notes	☒ Yes
Procedure recording	☒ Yes
Tooth chart	Not observed
Medical history	☒ Yes
Allergies	☒ Yes
Clinical attachments	Not observed
X-ray integration	Not observed
Treatment plan	☒ Yes
Follow-up tracking	☒ Yes

Gap: Medini CRM has no clinical documentation. This is P1 for doctor workflow.

10. Document Management

10.1 Document Types Observed

Type	Purpose
Consent form	Treatment consent
MC form	Medical certificate
Time off form	Staff leave
PDPA form	Data protection consent
X-ray	Radiographic imaging
Referral letter	Specialist referral

Gap: Medini CRM has no document management. This is P2.

11. Reporting

11.1 Reports Observed

Category	Reports
Patient	Daily visit, monthly visit, yearly visit, new patient, last treatment
Treatment	Daily, monthly, yearly, by doctor, by category
Financial	Revenue, payment, outstanding, deposit, panel, insurance
Operational	Feedback, incident, death, DOSH inspection
Inventory	Medicine, product, material (daily, monthly)
Staff	Commission, leave

Medini Status: Dashboard has KPIs but no detailed reports. P1.

12. Dashboard / KPI Comparison

Feature	Dr Partner	Medini CRM	Gap
Revenue tracking	☒ Yes	☒ Yes	—
Appointment count	☒ Yes	☒ Yes	—
Patient count	☒ Yes	☒ Yes	—
Outstanding	☒ Yes	☒ No	P0
Insurance claims	☒ Yes	☒ No	P0
Doctor performance	☒ Yes	☒ No	P1
Branch comparison	☒ Yes	☒ Yes	—
Real-time queue	☒ Yes	☒ No	P2
Treatment analytics	☒ Yes	☒ No	P1
Intelligence/AI	☒ No	☒ Yes	Medini advantage

13. User Roles

13.1 Roles Observed

Role	Access
HQ/Admin	Full access, all branches
Branch Manager	Own branch, reports, settings
Doctor	Own patients, clinical, appointments
Receptionist	Registration, appointments, payments
Finance	Payments, invoices, insurance, reports

Medini Status: Role model exists (Phase 3.1) but needs financial module separation.

14. Branch Management

14.1 Features Observed

Feature	Purpose
Branch edit	Update branch details
Branch logo	Branding
Staff assignment	Doctor/receptionist allocation
Tax setup	SST/tax configuration
Credit limit	Branch credit control

Medini Status: 14-branch model exists. Branch settings missing.

15. Communication

15.1 Features Observed

Feature	Purpose
WhatsApp	Patient messaging
SMS	Appointment reminders
Message templates	Pre-defined messages
Campaign	Bulk messaging
Frontdesk message	Internal notes

Medini Status: WhatsApp Hub exists (Phase 5). Message templates/campaigns missing.

16. Recall System

16.1 Workflow Observed

```
Treatment completed
↓
Recall interval set (e.g., 6 months)
↓
Recall due date calculated
↓
Reminder sent (SMS/WhatsApp)
↓
Patient contacted
↓
Appointment booked
↓
Recall completed
```

Medini Status: Recall Due exists (Phase 6). Recall workflow automation missing.

17. Financial Model

17.1 Observed Structure



17.2 Key Entities

Entity	Relationship
Patient	Has many invoices
Invoice	Has many payments
Payment	Belongs to invoice
Treatment	Billed on invoice
Panel/Insurance	Pays portion of invoice
Outstanding	Balance on invoice

Gap: Medini CRM has no financial entities. This is P0.

18. Medini Existing Capabilities

18.1 What Medini Already Has

Feature	Phase	Status
Dashboard Command Center	1-7	☑ Complete
Role-based access (4 roles)	3.1	☑ Complete
Branch scoping (14 branches)	1	☑ Complete
Patient List + Search + Filter	6	☑ Complete
Patient 360	6	☑ Complete
Family & Relationships	6.1	☑ Complete
Referral Network	6.1	☑ Complete
New Appointment	6.2	☑ Complete
New Patient Registration	6.3	☑ Complete
Shared Family Contact	6.3	☑ Complete
Follow-up workflow	5.1	☑ Complete
Dashboard reflection	6	☑ Complete
Financial isolation (RBAC)	3.1	☑ Complete
Responsive design	All	☑ Complete
Single HTML review artifact	All	☑ Complete

18.2 Medini Advantages Over Dr Partner

Feature	Medini	Dr Partner
Modern UI/UX	☑ Yes	✗ No (legacy)
Dashboard intelligence	☑ Yes	✗ No
Rule-based insights	☑ Yes	✗ No
Action-oriented workflow	☑ Yes	✗ No
Mobile responsive	☑ Yes	✗ No
Family relationship mapping	☑ Yes	✗ No
Shared contact handling	☑ Yes	✗ No
Real-time dashboard	☑ Yes	✗ No

19. Gaps — What Medini Is Missing

19.1 P0 — Critical (Must Have Before Production)

Gap	Why Critical	Module
Treatment catalog with codes/pricing	Cannot bill without treatments	Billing
Invoice generation	Legal/financial requirement	Finance
Payment processing	Revenue collection	Finance
Partial payment	Real-world payment behaviour	Finance
Outstanding tracking	Debt management	Finance
Insurance/Panel module	Corporate billing	Insurance
Receipt generation	Patient record	Finance
Allergies field	Clinical safety	Clinical
Medical history	Treatment contraindication	Clinical

19.2 P1 — Important (Should Have)

Gap	Why Important	Module
Clinical notes (SOAP)	Doctor documentation	Clinical
Treatment plan	Multi-visit planning	Clinical
Address field	Billing/recall	Patient
Email field	Digital communication	Patient
Doctor performance report	Productivity tracking	Reports
Treatment analytics	Business intelligence	Reports
Queue management	Front-desk operations	Appointments
Document upload	Record keeping	Documents

19.3 P2 — Useful (Nice to Have)

Gap	Why Useful	Module
Member/loyalty ID	Patient retention	Patient
Package/bundle	Marketing	Treatment
Voucher/discount	Promotions	Finance
Deposit tracking	Advance payment	Finance
Refund workflow	Customer service	Finance
Void payment	Error correction	Finance
Inventory basics	Stock tracking	Inventory
Staff commission	Payroll	HR

19.4 P3 — Optional (Defer)

Gap	Why Optional	Module
Occupation	Demographics only	Patient
Race/ethnicity	MOH reporting if needed	Patient
Equipment tracking	Maintenance	Operations
Death report	Regulatory	Reports
DOSH inspection	Compliance	Reports
Tourism report	Niche	Reports

20. Recommended Enhancements

20.1 Top 10 Recommendations

Rank	Feature	Why	Who Uses	Where	Priority	Dependencies
1	Treatment Catalog	Foundation for billing	All	Settings → Treatment	P0	Database schema
2	Invoice + Payment System	Revenue collection	Reception, Finance	Finance module	P0	Treatment catalog
3	Partial Payment + Outstanding	Real-world payment	Reception, Finance	Finance module	P0	Invoice system
4	Insurance/Panel Module	Corporate billing	Finance, Manager	Insurance module	P0	Invoice system
5	Allergies + Medical History	Clinical safety	Doctor, Reception	Patient 360	P0	Patient schema
6	Clinical Notes (SOAP)	Doctor documentation	Doctor	Patient 360 → Clinical	P1	Patient schema
7	Receipt Generation	Patient record	Reception	Finance module	P0	Payment system
8	Treatment Plan	Multi-visit tracking	Doctor	Patient 360 → Treatment	P1	Clinical notes
9	Address + Email Fields	Complete patient record	Reception	Patient registration	P1	Patient schema
10	Queue Management	Front-desk efficiency	Reception	Appointments	P1	Appointment system

20.2 Proposed Patient 360 Structure (Enhanced)

PATIENT 360
Patient Header (Name / MRN / Branch / Status)
Profile - Identity (Name, IC, DOB, Gender) - Contact (Phone, WhatsApp, Email, Address) - Allergies ⚠ (P0 – NEW) - Medical History (P0 – NEW)
Family & Relationships (Phase 6.1)
Referral Network (Phase 6.1)
Upcoming Appointment (Phase 6.2)
Treatment History (P1 – NEW) - Past treatments - Treatment plans - Clinical notes
Payments & Outstanding (P0 – NEW) - Invoices - Payment history - Partial payments - Outstanding balance
Insurance / Panel (P0 – NEW) - Active panels - Claims - Coverage
Documents (P2 – NEW) - Consent forms - X-rays - Referral letters
Timeline (Phase 6)
Notes (Phase 6)
Follow-up (Phase 5.1)

21. Proposed Appointment Domain (Future)

APPOINTMENT DOMAIN
Calendar View - Day / Week / Month - Doctor filter - Branch filter - Chair/room assignment
Queue Management Check-in - Waiting list Called/In-progress - Completed
Appointment Lifecycle - Booked → Confirmed → Checked-in → Waiting → In-progress → Completed - Cancelled / No-show tracking - Reschedule workflow
Treatment Link - Appointment → Treatment - Treatment → Invoice - Invoice → Payment
Recall Integration - Treatment completion → Recall scheduled - Recall due → Reminder sent - Recall booked → Appointment created

22. What Medini Should NOT Copy

Feature	Why Not
Legacy UI patterns	Medini has superior modern UI
Complex inventory	Overkill for dental CRM
Death/DOSH reports	Not relevant for dental
Machine/equipment tracking	Not core CRM function

23. Priority Roadmap

Phase 7 — Financial Foundation (P0)

- 1. Treatment catalog (codes, categories, pricing)
- 2. Invoice generation
- 3. Payment processing (full + partial)
- 4. Outstanding tracking
- 5. Receipt generation

Phase 8 — Insurance & Panel (P0)

- 1. Panel company setup
- 2. Insurance company setup
- 3. Claim workflow
- 4. Panel billing
- 5. Insurance payment tracking

Phase 9 — Clinical Safety (P0–P1)

- 1. Allergies field
- 2. Medical history
- 3. Clinical notes (simplified SOAP)
- 4. Treatment plan

Phase 10 — Operations (P1)

- 1. Queue management
- 2. Document upload
- 3. Address + email fields
- 4. Doctor performance reports
- 5. Treatment analytics

Phase 11 — Advanced (P2)

- 1. Package/bundle
- 2. Voucher/discount
- 3. Inventory basics
- 4. Staff commission
- 5. Recall automation

24. Conclusion

Medini CRM has a **superior dashboard and intelligence layer** compared to The Dr Partner. The Phase 1–7 Dashboard Command Center, Phase 6 Patient Management, and Phase 6.1–6.3 enhancements (Family, Referral, New Appointment, New Patient) are architecturally sound and visually modern.

However, Medini CRM **lacks the financial and clinical depth** required for real clinic operations:

- 1. **No treatment catalog** — cannot price or bill services
- 2. **No invoice/payment system** — cannot collect revenue
- 3. **No partial payment/outstanding** — cannot handle real-world payment behaviour
- 4. **No insurance/panel** — cannot bill corporate clients
- 5. **No allergies/medical history** — clinical safety risk

Recommendation: Before building production backend, implement P0 gaps (treatment catalog, invoice/payment, outstanding, insurance/panel, allergies) as prototype modules in the Single HTML review artifact. This ensures the UX workflow is validated before database/API development.

25. Evidence Log

Source	What Was Observed	Why It Matters	Medini Recommendation
Dr Partner → Setting → Treatment	Treatment category + setup pages	Treatment catalog structure	Add treatment catalog with codes/pricing
Dr Partner → Payment	Payment collection page	Payment workflow	Add invoice + payment system
Dr Partner → Outstanding	Outstanding invoice page	Debt tracking	Add outstanding balance tracking
Dr Partner → Unpaid	Unpaid invoice list	Partial payment model	Add partial payment support
Dr Partner → Panel	Panel invoice/payment pages	Corporate billing	Add insurance/panel module
Dr Partner → Patient	Patient registration fom	Patient fields	Add allergies, medical history, address
Dr Partner → SOAP	SOAP option setup	Clinical documentation	Add simplified clinical notes
Dr Partner → Reports	20+ report pages	Business intelligence	Add treatment/doctor performance reports
Dr Partner → Queue	Queue page with polling	Front-desk operations	Add queue management
Dr Partner → POS	Point of sale	Payment processing	Integrate POS with invoice

Privacy Check: ☒ PASS — No real patient names, IC numbers, phone numbers, addresses, clinical notes, X-rays, insurance member numbers, or payment account details included in this report.